KHUTSO MASHAPU

Johannesburg, Gauteng

076-433-3580 | khutsomashapu663@gmail.com | GitHub: https://github.com/Diego23-co

SUMMARY

Computer Science Honours candidate specialising in Machine Learning and Predictive Analytics, with hands-on experience building end-to-end ML solutions using Python and Scikit-learn. Proven ability in data preprocessing, feature engineering and model evaluation, with active research in sports outcome prediction and applied ML on real-world datasets.

EDUCATION

University of Witwatersrand *Johannesburg, South Africa*

BSc. Computer Science & Mathematics **Feb 2022-Dec 2025**

BSc. Hons Computer Science **Feb 2026-Dec 2026** *(expected)*

RELEVANT COURSEWORK

- Machine Learning
- Software Engineering
- LLMs
- Data Structures and Algorithms
- Artificial Intelligence
- Database Management Systems
- Data Science
- Operating Systems

EXPERIENCE

Freelynk - Freelancer Platform | JavaScript, Node.js, Firebase, Microsoft Azure **May 2025**

- Led a five-person team to develop a web-based Freelancer platform aimed at connecting clients with service providers.
- Adopted Agile methodology, integrated CI/CD principles, and followed a test-driven development (TDD) approach throughout the development lifecycle.
- Coordinated 6 weekly meetings and delivering a final presentation to 50+ stakeholders resulting in a 5/5 client satisfaction score.

GoPredict Web Platform | Python, Flask, SQL, Git & Github **Dec 2025**

- Built a full-stack web app integrating a live sports API to display real-time match data with user prediction functionality.
- Managed end-to-end development including SQL database design, Flask backend and deployment, with strong potential for user engagement.

SA Crime Count Prediction | Python, Scikit-learn, Pandas, Matplotlib, Github **Feb 2026**

- Built and deployed an end-to-end ML pipeline predicting South African provincial crime counts via a live REST API.
- Achieved Test R^2 of 0.997 with Random Forest, outperforming Linear Regression by 28%.
- Live at sa_crime_prediction.onrender.com

Boxing Match Outcome Prediction| Data Science & Machine Learning Academic Research **Mar 2026**

- Conducting academic research comparing ML model predictions against human judge and referee decisions in professional boxing matches.

SKILLS

Programming: Python, SQL
Machine Learning & AI: scikit-learn, NumPy, pandas, model evaluation, feature engineering, predictive modelling
Data & Analytics: Data preprocessing, data visualization, statistical analysis
Software & Tools: Git, Github, Jupyter Notebook, Vs Code, Linux, Flask, Render